

# GS-441524 Treatment in a Cat With Feline Infectious Peritonitis Virus and Pyogranulomatous Transverse Colon Lesion

TAE-YOUNG KIM<sup>1,2</sup> and YE-IN OH<sup>2</sup>

<sup>1</sup>Jukjeon Animal Medical Center, Daegu, Republic of Korea;

<sup>2</sup>Department of Veterinary Internal Medicine, College of Veterinary Medicine and Institute for Veterinary Biomedical Science, Kyungpook National University, Daegu, Republic of Korea

## Abstract

**Background/Aim:** Feline infectious peritonitis (FIP) is a fatal disease caused by feline coronavirus (FCoV), manifesting as effusive (wet) or non-effusive (dry) forms. Granulomatous lesions in the gastrointestinal tract, particularly the colon, are rare and pose diagnostic and therapeutic challenges.

**Case Report:** A 7-year-old castrated male domestic shorthair cat (4.1 kg) presented with anorexia and vomiting. Blood tests showed an albumin:globulin ratio of 0.5, a mild elevation in aspartate aminotransferase, mild leukocytosis, and a severe elevation of feline serum amyloid A. The abdominal ultrasound and computed tomography imaging showed a mass in the transverse colon and surrounding mesenteric lymph nodes. The mass was observed to be relatively well vascularized, with areas of low-density uneven necrosis. Histological examination revealed severe pyogranulomatous inflammation with macrophages, neutrophils, and lymphocytes predominantly present in both the affected colon and lymphoid tissues. Immunohistochemistry for feline infectious peritonitis (FIP) virus antigen of the lesion showed a strong positive result, confirming limited and localized lesion induction by FIP virus infection. The cat received GS-441524 for 12 weeks, resulting in clinical improvement, lesion resolution, and normalized lab results. RT-PCR and antibody tests were negative post-treatment.

**Conclusion:** This case report describes a rare presentation of FIP virus infection, characterized by localized lesions confined to a specific segment of the colon. In this case, treatment with the nucleoside analog GS-441524 was very effective in improving the localized lesion and demonstrated excellent efficacy in clearing the FIP virus.

**Keywords:** Feline infectious peritonitis, GS-441524, nucleoside analog, pyogranulomatous inflammation.



Ye-In Oh, Department of Veterinary Internal Medicine, College of Veterinary Medicine, Kyungpook National University, 41566, Daegu, Republic of Korea. E-mail: imyein5@knu.ac.kr

Received April 7, 2025 | Revised May 19, 2025 | Accepted May 21, 2025



This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

©2025 The Author(s). Anticancer Research is published by the International Institute of Anticancer Research.

## Introduction

Feline infectious peritonitis (FIP) is a fatal and complex disease caused by feline coronavirus (FCoV) that affects cats worldwide (1, 2). This disease primarily manifests in two types: the effusive (wet) type, characterized by accumulation of fluid in body cavities, and the non-effusive (dry) type, marked by granulomatous lesions in various organs (2-6). Granulomatous inflammation, a characteristic of non-effusive FIP, involves the formation of granulomas surrounded by neutrophils, macrophages, and lymphocytes. Although these lesions can occur in various organs, their presence in the gastrointestinal tract, particularly in the colon, is rare and presents a therapeutic challenge (7-9).

FIP has long been considered a fatal disease in affected cats; until recently, no effective antiviral treatments were available. Treatment did not reduce the high mortality rate (>95%) of FIP or slow disease progression (4, 8, 10). The nucleoside analog GS-441524 has emerged as a promising antiviral agent against FIP (11, 12). GS-441524 is the parent nucleoside of remdesivir, which is used to treat Ebola and coronavirus disease (COVID-19) (13). GS-441524 inhibits viral RNA-dependent RNA polymerase, thereby interfering with viral replication (11, 12).

This case report describes GS-441524 treatment for a granulomatous lesion in the transverse colon of a cat infected with the FIP virus.

## Case Report

A 7-year-old neutered male domestic shorthair cat presented with anorexia and intermittent vomiting, characterized by small, frequent episodes occurring once weekly. Physical examination revealed mild dehydration with no other significant abnormalities. Informed consent was obtained from the owner of the animal described in this work for all procedures undertaken.

Blood tests showed a low albumin-to-globulin (A:G) ratio (0.5), mildly elevated aspartate aminotransferase (AST) level (78 U/l; reference range=0-48 U/l), mild leukocytosis ( $19.84 \times 10^9$ /l; reference range= $2.87$ - $17.02 \times 10^9$ /l), and

significantly elevated feline serum amyloid A (fSAA) level (19.9 µg/ml; reference range <5.0 µg/ml). The cat tested negative for feline leukemia virus antigen and feline immunodeficiency virus antibody kit [IDEXX Snap Combo kit (IDEXX Laboratories, Inc., Westbrook, MN, USA)].

Abdominal ultrasonography revealed loss of the wall layer structure and thickening of the colonic wall in the transverse colon region (Figure 1A and B). The wall thickness measured 8.4 mm and 5.9 mm, and no significant physical obstruction of the intestinal lumen was noted. The adjacent mesenteric lymph node showed an increase in size to  $11.9 \times 38.6$  mm, with mass formation and morphological changes, accompanied by increased echogenicity of the surrounding peritoneal fat (Figure 1C). Abdominal computed tomography (CT) imaging confirmed the abnormal thickening and increased radiodensity of the transverse colonic wall (Figure 1D). At the mid-abdominal level in the transverse plane, three contiguous atypical masses originating from the mesenteric lymph node were observed, adjacent to the transverse colonic wall. These adjacent tumors exhibited relatively well-developed vasculature, and post-contrast portal phase imaging showed heterogeneous enhancement, including areas of low density within the internal parenchyma (Figure 1E). Peritoneal fat stranding around the lesion indicated abnormal changes and inflammation in the adipose tissue and surrounding structures.

Cytological examination conducted via ultrasound-guided fine-needle aspiration of the lesion showed low cellularity, with no evidence of neoplastic cells suggesting malignancy (Figure 2A). Cytology of the mesenteric lymph node revealed a mixture of epithelial cells, mesenchymal cells, and clusters of adipocytes, with an increase in background lipid vacuoles, indicating nonspecific non-infectious inflammatory lesions with some fat necrosis (Figure 2B). Histopathological examination of the affected colon and lymphatic tissue revealed severe suppurative granulomatous inflammation, primarily composed of macrophages, neutrophils, and lymphocytes (Figure 2C-E). This inflammation extended throughout the colonic wall, resulting in significant thickening (Figure 2C). No causative

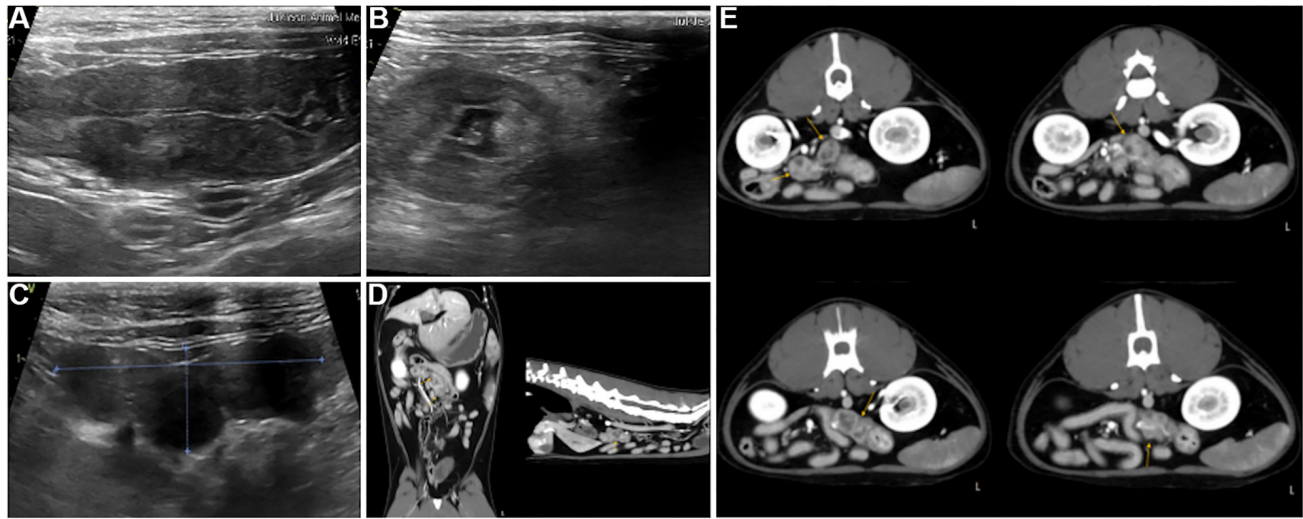


Figure 1. Abdominal ultrasonography of the cat. A, B) Loss of the intestinal wall layering and increased thickness of part of transverse colon. C) Increased echogenicity of the peritoneal fat surrounding the colonic wall lesion and the adjacent mesenteric lymph node enlargement. Computed tomography of the cat. D) Abnormal thickening and increased radiodensity of the transverse colonic wall (yellow arrow). E) At the mid-abdominal level in the transverse plane, three contiguous atypical masses originating from the mesenteric lymph node were observed. The internal parenchyma showed heterogeneous enhancement with areas of low density (yellow arrow).

agents were identified on Hematoxylin and Eosin (H&E) staining. Fungal pathogens were not observed in the inflamed colonic wall with Periodic Acid-Schiff (PAS) staining (Figure 2F), nor were *Mycobacterium* spp. detected with Fite staining (Figure 2G). Immunohistochemical staining for FIP virus antigen showed strong positive reactivity in the inflammatory lesions along the colonic wall (Figure 2H and I). Based on these diagnostic findings, the localized lesions along the transverse colonic wall and the associated lymphadenopathy and inflammatory response were diagnosed as being caused by FIP virus infection.

The patient was treated with GS-441524 (6 mg/kg, q24h, SC) for 12 weeks. Symptomatic treatment included a broad-spectrum antibiotic (Amoxicillin-Clavulanate, 62.5 mg/cat, q12h, PO), prednisolone (0.25 mg/kg, q12h, PO), an antiemetic (maropitant 1 mg/kg, q12h, PO), and a proton pump inhibitor (omeprazole 5 mg/cat, q24h, PO). Gastrointestinal clinical signs improved within 3 days of starting treatment, and symptomatic medications were discontinued after seven days.

At the time of the initial biopsy, the cat's weight was approximately 4 kg, but it decreased to 3.5 kg before starting GS-441524 treatment. After treatment commenced, the cat's weight steadily increased, reaching 4.6 kg by the end of the 12-week treatment period, and subsequently increasing to 5.72 kg at follow-up. The episodes of vomiting ceased. The A:G ratio was low (0.5) at the time of biopsy and at the start of GS-441524 treatment but increased gradually, reaching 0.7 by the end of the 12-week treatment and later rising to 1. The fSAA levels initially were 19.9 µg/ml and increased to 25.5 µg/ml at the start of GS-441524 treatment. However, with continued treatment, fSAA levels decreased to within the normal range (<5.0 µg/ml) by the end of the treatment period and remained normal at the final follow-up. By the end of the treatment, the localized lesions in the transverse colon and the surrounding lymphadenopathy had resolved (Figure 3A). The colonic wall maintained its normal structure at follow-ups at 86- and 308 weeks post-treatment (Figure 3B and C). Post-treatment rapid testing for FCoV antibodies was negative, and RT-PCR testing for FCoV antigens in whole blood and feces also returned negative

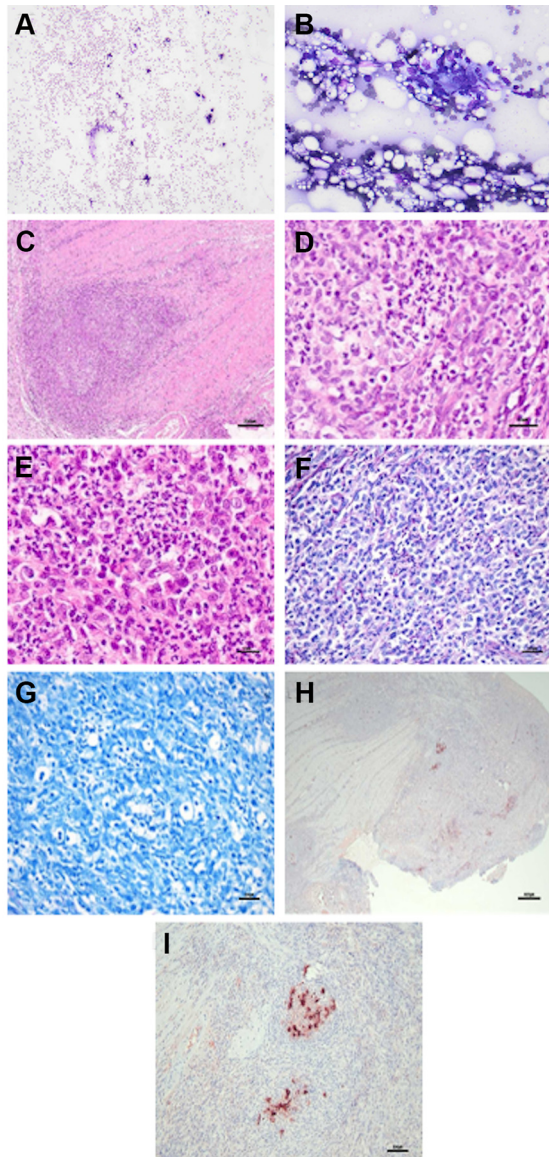


Figure 2. Cytology of the obtained colonic lesion [Diff-Quik staining, 100× (A)] and adjacent enlarged mesenteric lymph node [Diff-Quik staining, 400× (B)]. A, B) The cellularity was relatively low, and no significant evidence of malignant neoplasia is observed. The findings suggested a nonspecific, non-infectious inflammatory lesion with some fat necrosis. Histopathology of the obtained colonic lesion [H&E staining, 100× (C), 400× (D), PAS staining, 200× (F), Fite staining, 200× (G), Immunohistochemistry, 50× (H), 200× (I)] and the adjacent enlarged mesenteric lymph node [H&E stain, 400× (E)]. D, E) Severe suppurative granulomatous inflammation is observed, predominantly composed of macrophages, neutrophils, and lymphocytes. This inflammation extends throughout the entire colonic wall (C), resulting in significant thickening. F, G) Fungal pathogens and *Mycobacterium* spp. are not observed in the inflamed areas of the thickened colonic wall. H, I) Strong positive reactivity for the feline infectious peritonitis virus is confirmed along the inflamed areas of the thickened colonic wall.

results. Serum A:G ratios and fSAA levels remained within normal ranges. The patient has survived up to 44 months post-diagnosis with regular follow-up.

## Discussion

This case highlights the potential of GS-441524 as an effective therapeutic agent for feline infectious peritonitis (FIP), even in atypical presentations. The nonspecific initial clinical signs could mimic various gastrointestinal disorders, but blood test abnormalities and imaging findings indicated the systemic nature of the disease. Abdominal ultrasonography and CT provided essential insights into colonic wall abnormalities and mesenteric lymph node inflammation, aiding diagnosis (6, 11).

FIP remains a complex and often fatal feline disease due to diagnostic challenges and limited treatment options. It can affect various organs and present in diverse types (3, 6), but localized intestinal lesions are rare (7). The development of these lesions may involve a partial cell-mediated immune response that traps the virus in intestinal macrophages without eliminating it, causing chronic granulomatous inflammation (7, 14).

Atypical FIP is marked by localized granulomatous lesions, often in the distal small intestine, ileocecal junction, or colon, along with adjacent lymphadenopathy (7, 14, 15). Such lesions can resemble tumors or infections like mycobacteriosis, necessitating histopathological examination for accurate diagnosis (7, 14, 16). In this case, localized lesions were restricted to the transverse colonic wall, characterized by suppurative granulomatous inflammation. This case underscores the importance of histopathology in distinguishing FIP from other diseases.

In a previous study, intestinal changes associated with FIP on abdominal ultrasonography included wall thickening, which was observed in 19% (3/16) of cases (17). The highest rate of ultrasonographic changes was noted in the lymph nodes, observed in 75% (12/16) of cases. In another study, atypical FIP was observed in 17% (26/156) of cases (7). The lesions were located in the distal small intestine, including the ileum, ileocecal





Figure 3. Abdominal ultrasonography of the cat. A) At the end of the 12-week treatment period, the localized lesions in the transverse colon and the surrounding lymph nodes had all improved. B) At the 86-week follow-up post-treatment, and C) at the 308-week follow-up post-treatment, the colonic wall maintained its normal structure.

junction, and colon, and the adjacent lymph nodes were also affected. These findings highlight the diverse presentations of FIP and the importance of thorough diagnostic imaging to identify atypical cases involving the gastrointestinal tract. The present case was classified as atypical FIP, with the adjacent lymph nodes affected. Notably, the lesions were localized only in specific areas of the transverse colonic wall, which could have led to a missed diagnosis. Therefore, careful diagnostic imaging is crucial to ensure no lesions are overlooked.

The treatment regimen centered on GS-441524 was proven to be highly effective, as evidenced by the rapid improvement in clinical signs and the resolution of localized lesions. The successful outcome in this case supports the growing evidence for the use of GS-441524 in FIP treatment. The cat's significant weight gain, along with normalization of the A:G ratio and fSAA level, highlights the drug's efficacy in reducing systemic inflammation and promoting recovery. The sustained negative results for FCoV antibodies and antigen after treatment indicate successful virus eradication, reinforcing the therapeutic potential of GS-441524.

This case underscores the importance of a comprehensive diagnostic approach that includes clinical signs, laboratory findings, imaging, and histopathological examinations to accurately diagnose and manage FIP. The successful resolution of this rare colonic manifestation suggests that GS-441524 may be a viable option for cats with atypical FIP presentations.

No recurrence was observed during the three-year follow-up, demonstrating the drug's long-term effectiveness. While GS-441524 shows promise, further research is needed to investigate its long-term effects and explore alternative nucleoside analogs for FIP treatment.

### Conflicts of Interest

The Authors declare that they have no competing interests in relation to this study.

### Authors' Contributions

Conceptualization, Investigation, Data curation, Writing – Original Draft, Visualization Writing-Original Draft: TY Kim; Conceptualization, Data curation, Writing – Review & Editing, Supervision: Oh YI.

### Funding

The Authors received no financial support for the research, authorship, and/or publication of this article.

### Artificial Intelligence (AI) Disclosure

No artificial intelligence (AI) tools, including large language models or machine learning software, were used in the preparation, analysis, or presentation of this manuscript.

## References

- 1 Pedersen NC: A review of feline infectious peritonitis virus infection: 1963-2008. *J Feline Med Surg* 11(4): 225-258, 2009. DOI: 10.1016/j.jfms.2008.09.008
- 2 Rohrbach BW, Legendre AM, Baldwin CA, Lein DH, Reed WM, Wilson RB: Epidemiology of feline infectious peritonitis among cats examined at veterinary medical teaching hospitals. *J Am Vet Med Assoc* 218(7): 1111-1115, 2001. DOI: 10.2460/javma.2001.218.1111
- 3 Addie D, Belák S, Boucraut-Baralon C, Egberink H, Frymus T, Gruffydd-Jones T, Hartmann K, Hosie MJ, Lloret A, Lutz H, Marsilio F, Pennisi MG, Radford AD, Thiry E, Truyen U, Horzinek MC: Feline infectious peritonitis. ABCD guidelines on prevention and management. *J Feline Med Surg* 11(7): 594-604, 2009. DOI: 10.1016/j.jfms.2009.05.008
- 4 Dickinson PJ, Bannasch M, Thomasy SM, Murthy VD, Vernau KM, Liepnieks M, Montgomery E, Knickelbein KE, Murphy B, Pedersen NC: Antiviral treatment using the adenosine nucleoside analogue GS-441524 in cats with clinically diagnosed neurological feline infectious peritonitis. *J Vet Intern Med* 34(4): 1587-1593, 2020. DOI: 10.1111/jvim.15780
- 5 Felten S, Hartmann K: Diagnosis of feline infectious peritonitis: a review of the current literature. *Viruses* 11(11): 1068, 2019. DOI: 10.3390/v11111068
- 6 Kipar A, May H, Menger S, Weber M, Leukert W, Reinacher M: Morphologic features and development of granulomatous vasculitis in feline infectious peritonitis. *Vet Pathol* 42(3): 321-330, 2005. DOI: 10.1354/vp.42-3-321
- 7 Harvey CJ, Lopez JW, Hendrick MJ: An uncommon intestinal manifestation of feline infectious peritonitis: 26 cases (1986–1993). *J Am Vet Med Assoc* 209(6): 1117-1120, 1996. DOI: 10.2460/javma.1996.209.06.1117
- 8 Pedersen NC: An update on feline infectious peritonitis: diagnostics and therapeutics. *Vet J* 201(2): 133-141, 2014. DOI: 10.1016/j.tvjl.2014.04.016
- 9 Van Kruiningen HJ, Ryan MJ, Shindel NM: The classification of feline colitis. *J Comp Pathol* 93(2): 275-294, 1983. DOI: 10.1016/0021-9975(83)90015-4
- 10 Sherding RG: Feline infectious peritonitis (feline coronavirus). *Saunders Manual of Small Animal Practice*: 132-143, 2006. DOI: 10.1016/B0-72-160422-6/50012-7
- 11 Barua S, Kaltenboeck B, Juan YC, Bird RC, Wang C: Comparative evaluation of GS-441524, teriflunomide, ruxolitinib, molnupiravir, ritonavir, and nirmatrelvir for in vitro antiviral activity against feline infectious peritonitis virus. *Vet Sci* 10(8): 513, 2023. DOI: 10.3390/vetsci10080513
- 12 Murphy BG, Perron M, Murakami E, Bauer K, Park Y, Eckstrand C, Liepnieks M, Pedersen NC: The nucleoside analog GS-441524 strongly inhibits feline infectious peritonitis (FIP) virus in tissue culture and experimental cat infection studies. *Vet Microbiol* 219: 226-233, 2018. DOI: 10.1016/j.vetmic.2018.04.026
- 13 Malin JJ, Suárez I, Priesner V, Fätkenheuer G, Rybníček J: Remdesivir against COVID-19 and Other Viral Diseases. *Clin Microbiol Rev* 34(1): e00162-20, 2020. DOI: 10.1128/CMR.00162-20
- 14 Hartmann K: Feline infectious peritonitis. *Vet Clin North Am Small Anim Pract* 35(1): 39-79, vi, 2005. DOI: 10.1016/j.cvsm.2004.10.011
- 15 Addie DD: Feline coronavirus infections. In: *Infectious diseases of the dog and cat*, 4e. Greene CE (eds.). St. Louis, MO, USA, Saunders Elsevier, pp. 92-108, 2012.
- 16 Kipar A, Koehler K, Bellmann S, Reinacher M: Feline infectious peritonitis presenting as a tumour in the abdominal cavity. *Vet Rec* 144(5): 118-122, 1999. DOI: 10.1136/vr.144.5.118
- 17 Lewis KM, O'Brien RT: Abdominal ultrasonographic findings associated with feline infectious peritonitis: a retrospective review of 16 cases. *J Am Anim Hosp Assoc* 46(3): 152-160, 2010. DOI: 10.5326/0460152